

MSc thesis in Geomatics

Deep Learning Classification Leveraging Multi-Temporal Historical Maps for Enabling Urban Analysis in The Past

Andika Hadi Hutama

June 2026

A thesis submitted to the Delft University of Technology in partial fulfillment of
the requirements for the degree of Master of Science in Geomatics

Andika Hadi Utama: *Deep Learning Classification Leveraging Multi-Temporal Historical Maps for Enabling Urban Analysis in The Past* (2026)

© This work is licensed under a Creative Commons Attribution 4.0 International License. To view a copy of this license, visit <http://creativecommons.org/licenses/by/4.0/>.

The work in this thesis was carried out in the:



Geo-Database Management Centre
Delft University of Technology

Supervisors: Dr.ir. B.M. (Martijn) Meijers
Prof.dr. A.R. (Ana) Pereira Roders

Co-reader: TBC

Abstract

i Info

Should fit on one page.

Lemongrass frosted gingerbread bites banana bread orange crumbled lentils sweet potato black bean burrito green pepper springtime strawberry ginger lemongrass agave green tea smoky maple tempeh glaze enchiladas couscous. Cranberry spritzer Malaysian cinnamon pineapple salsa apples spring cherry bomb bananas blueberry pops scotch bonnet pepper spiced pumpkin chili lime eating together kale blood orange smash arugula salad. Bento box roasted peanuts pasta Sicilian pistachio pesto lavender lemonade elderberry Southern Italian citrusy mint lime taco salsa lentils walnut pesto tart quinoa flatbread sweet potato grenadillo.

Lemongrass frosted gingerbread bites banana bread orange crumbled lentils sweet potato black bean burrito green pepper springtime strawberry ginger lemongrass agave green tea smoky maple tempeh glaze enchiladas couscous. Cranberry spritzer Malaysian cinnamon pineapple salsa apples spring cherry bomb bananas blueberry pops scotch bonnet pepper spiced pumpkin chili lime eating together kale blood orange smash arugula salad. Bento box roasted peanuts pasta Sicilian pistachio pesto lavender lemonade elderberry Southern Italian citrusy mint lime taco salsa lentils walnut pesto tart quinoa flatbread sweet potato grenadillo.

Acknowledgements

Thanks to everyone, especially my supervisors and my dog (in that order). Obviously, thanks also to the brilliant minds who created this *fantastic* template!

Lemongrass frosted gingerbread bites banana bread orange crumbled lentils sweet potato black bean burrito green pepper springtime strawberry ginger lemongrass agave green tea smoky maple tempeh glaze enchiladas couscous. Cranberry spritzer Malaysian cinnamon pineapple salsa apples spring cherry bomb bananas blueberry pops scotch bonnet pepper spiced pumpkin chili lime eating together kale blood orange smash arugula salad. Bento box roasted peanuts pasta Sicilian pistachio pesto lavender lemonade elderberry Southern Italian citrusy mint lime taco salsa lentils walnut pesto tart quinoa flatbread sweet potato grenadillo.

Thai super chili apricot salad cocoa dark chocolate vitamin glow mushroom risotto red amazon pepper simmer udon noodles soba noodles dragon fruit cherries strawberry mango smoothie basil chickpea crust pizza cauliflower cherry bomb pepper mediterranean street style Thai basil tacos. Balsamic vinaigrette Indian spiced kimchi tofu sandwiches smoked tofu apple vinaigrette salty Thai sun pepper cayenne four-layer fiery fruit peach strawberry mango vegan Bulgarian carrot Italian linguine puttanesca green bowl lemon red lentil soup overflowing berries habanero golden one bowl.

Contents

1. Introduction	2
2. Related Work	4
3. Related Work	6
4. Related Work	8
5. Result and Analysis	10
6. Conclusion	12
6.1. Conclusion	12
6.2. Discussion	12
6.3. Future Work	12
A Declaration of AI/LLM usage	14
B Reproducibility self-assessment	16
B.1 Example statement	16
B.2 Examples of good MSc Geomatics repositories	16
C Some UML diagrams	18
Bibliography	20

List of figures

Figure C.1 The UML diagram of something that looks important.	18
--	----

List of tables

List of algorithms

TODOs

1. Introduction

2. Related Work

3. Related Work

4. Related Work

5. Result and Analysis

6. Conclusion

6.1. Conclusion

6.2. Discussion

6.3. Future Work

6.3. *Future Work*

A Declaration of AI/LLM usage

If you have used Artificial Intelligence (AI), Large Language Models (LLMs)—such as Claude, ChatGPT, Gemini, Copilot, or similar tools—at any stage of your MSc thesis, please explicitly disclose this in your document. Include the following details:

1. **What:** Specify which AI/LLM tool(s) you used (eg ChatGPT, Claude, Copilot, etc.).
2. **How:** Briefly describe how you used these tools (eg generating outlines, writing, coding, data analysis, editing, etc.).
3. **Extent:** Indicate the extent of their use (eg occasional assistance, regular drafting, checking grammar, etc.).
4. **Ethics:** Confirm adherence to academic integrity and that AI/LLM did not generate falsified, fictional, or plagiarized content.

Example statement: “Some parts of this thesis benefited from the use of AI tools such as ChatGPT for initial text drafts and grammar suggestions. All results, analyses, and conclusions are my own, and I verified that the content is original and meets academic integrity standards.”

B Reproducibility self-assessment

To promote open science and research integrity, please include a statement about the reproducibility of your MSc thesis work.

Address the following points concisely:

1. **Data Availability:** Is your data (that you used as input, or that you generated) publicly accessible? If yes, provide a persistent link (ideally a DOI, you can archive our datasets on GitHub or in the 4TU repository); if no, briefly explain why not (eg privacy, licensing).
2. **Code Availability:** Is your code and analysis workflow openly available? This can be for instance via GitHub, OSF, Zenodo. Include the relevant link or DOI if applicable. It is suggested to follow those guidelines for reproducibility: <https://geomatics.bk.tudelft.nl/geogeeks/git/goodgit/>.
3. **Documentation:** Have you provided clear instructions, README files, or notebooks so others can understand and reproduce your results?
4. **Reproducibility Rating:** Self-rate your overall reproducibility using the following scale:
 - **High:** All data and code fully available, well-documented, and tested to reproduce main results.
 - **Moderate:** Most data and code available, basic documentation, some steps may require clarification.
 - **Low:** Limited or no data/code sharing, minimal documentation, reproduction unlikely.

B.1 Example statement

"All data and analysis code related to this thesis are openly accessible at [insert link]. Complete instructions and a sample workflow are provided in the project README. I rate the reproducibility of this thesis as **high** according to the provided scale."

B.2 Examples of good MSc Geomatics repositories

1. <https://github.com/chenzhaiyu/points2poly>
2. <https://github.com/ellieroy/no-floors-inference-NL>
3. <https://github.com/NoortjevanderHorst/treegrowthmodelling>
4. <https://github.com/fabisser/stylesdf>

C Some UML diagrams

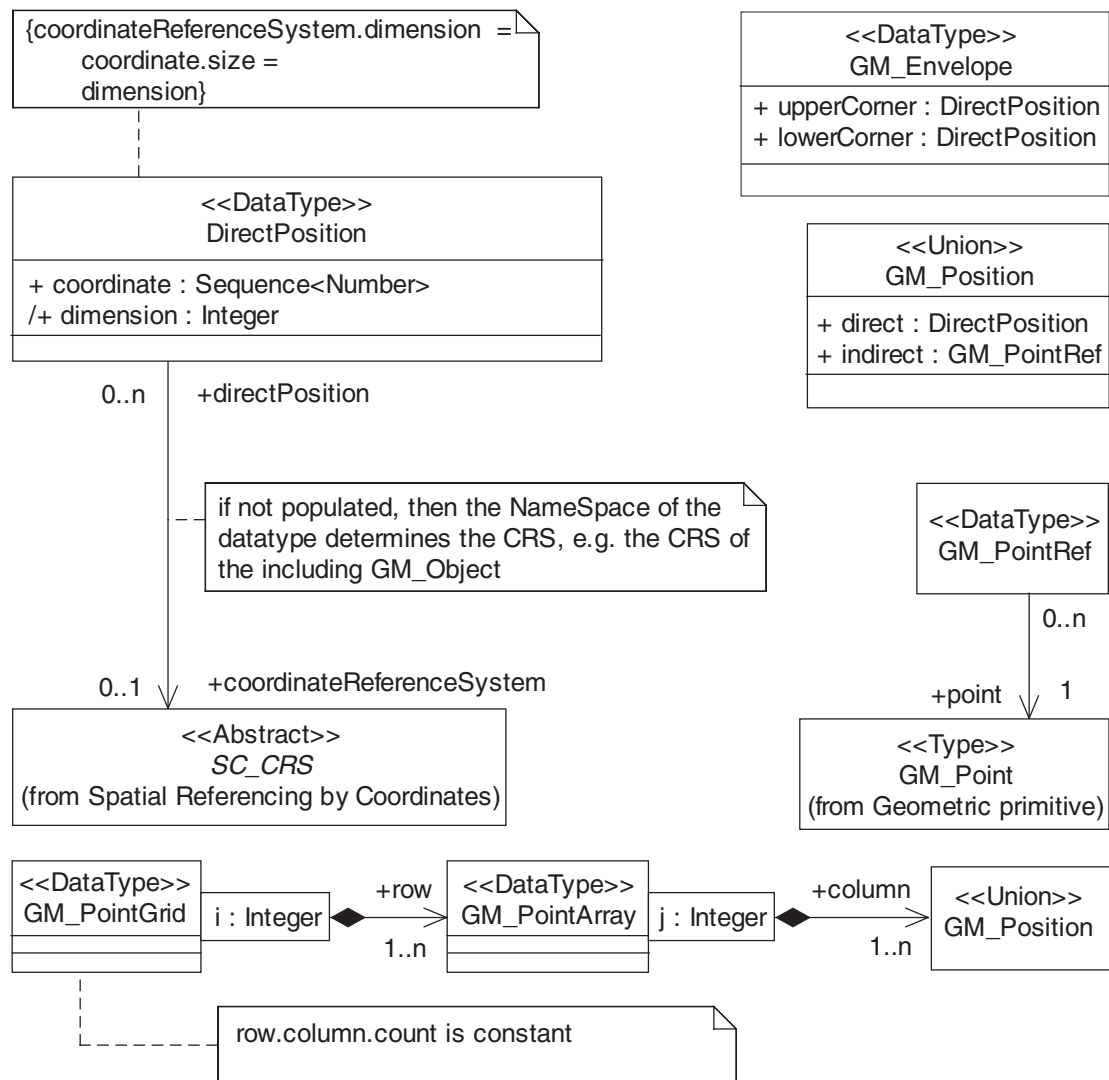


Figure C.1: The UML diagram of something that looks important.

Bibliography

